

# Synthetic Minds in Normative Sandboxes: Exaptation, Spandrels, and Multilevel Selection in Agent-Based Legal Simulation

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## Abstract

This paper introduces the Synthetic Minds Normative Sandbox (SNMS), a second-generation agent-based simulation framework for modeling the emergence and transformation of legal norms within multi-agent institutional environments. SNMS advances beyond first-generation normative automata by equipping agents with cognitively differentiated MindProfiles — composite structures encoding cognitive, affective, and doctrinal dimensions — calibrated empirically to eight Argentine institutional archetypes. Central to the architecture is the integration of TribeV2 (King, Benchetrit, Rapin, Brooks, Begany, Raugel & Banville, 2026), a tri-modal foundation model of human brain activity trained on over 1,000 hours of fMRI data across 720 subjects. In SNMS, the cortical encoding vectors produced by TribeV2 over normative text excerpts serve as a proxy for differential cognitive load across archetypes. The theoretical framework draws on three pillars: (1) Gould & Vrba's (1982) exaptation; (2) Gould & Lewontin's (1979) spandrels; and (3) Wilson & Sober's (1994) multilevel selection. Four novel metrics operationalize the framework: the Functional Drift Index (FDI), the Spandrel Emergence Metric (SEM), the Multilevel Selection Ratio (MSR), and the Mean Compliance Index (MCI). A proof of concept applied to a simulated Argentine labor reform scenario across 100 agents, 20 rounds, and 5 random seeds yields a Compliance Legitimacy Score (CLS) of 0.89, an FDI of 0.42, two normative spandrels by round 14, and an MSR of 0.73.

Keywords: normative simulation, agent-based modeling, exaptation, spandrels, multilevel selection, TribeV2, cognitive load, Argentine labor law, extended phenotype theory, evolutionary game theory

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## 1. Introduction

### 1.1 Epistemological Positioning

SNMS operates within the Lakatosian framework of scientific research programs (Lakatos, 1978): the hard core is the claim that cognitive heterogeneity among institutional actors produces qualitatively distinct normative dynamics relative to parameterized homogeneity. The protective belt consists of the MindProfile architecture, the TribeV2 integration, and the four metrics. The program is progressive insofar as it generates novel falsifiable predictions (Section 4.2) that its predecessor (Lerer, 2026a) does not entail. Following Hull (1988), the unit of selection in this research program is not the individual norm but the archetype-norm interaction pattern — the entity whose differential fitness drives normative evolution.

## 1.2 The Problem with First-Generation Normative Simulators

Agent-based models (ABMs) have become a standard tool in computational social science for studying the emergence, diffusion, and collapse of social norms (Axelrod, 1986; Epstein, 2006; Tverskoi, Gavrillets & Sánchez, 2024). Despite these advances, a structural limitation persists across virtually all existing normative simulators: agents are treated as parameterized automata. The cognitive architecture through which real institutional actors — judges, legislators, union leaders, informal workers — interpret, contest, and instrumentalize legal norms remains entirely absent. The cognitive distance between archetypes is not merely quantitative: it is qualitative — different doctrinal priors, different affective valences, different capacities for abstract norm processing. When simulation models collapse this heterogeneity into a single scalar, they sacrifice the very dynamics they purport to model.

## 1.3 TribeV2 as a Foundation for Cognitive Differentiation

The release of TribeV2 (King et al., 2026) opens a novel methodological pathway. TribeV2 is a tri-modal foundation model trained on over 1,000 hours of fMRI recordings from 720 subjects, capable of predicting high-resolution brain responses to novel stimuli with zero-shot generalization across subjects, tasks, and languages. It recovers well-established neuro-linguistic phenomena in silico — Broca's lateralization, the Visual Word Form Area, syntactic responses in the superior temporal gyrus — and was designed precisely for in-silico experimentation of the kind proposed here. As King et al. (2026) state explicitly, TribeV2 enables 'tested on seminal visual and neuro-linguistic paradigms, it recovers a variety of results established by human neuroscience experiments.'

In SNMS, TribeV2's language encoding pathway provides a proxy for cognitive load induced by normative text. The PoC reported here uses theoretically assigned MindProfile parameters inspired by TribeV2's cognitive load logic (see Section 3.2 for the explicit architecture and Section 5.3 for honest acknowledgment of this limitation). Full TribeV2 API integration — passing actual norm text excerpts through the encoder at each round — is the immediate next step and constitutes the primary future work.

## 1.4 Research Question and Contributions

The central research question: Does the introduction of cognitively differentiated MindProfiles produce qualitatively distinct emergent norm dynamics in agent-based legal simulation relative to parameterized automata? Four contributions: (1) the SNMS architecture as second-generation normative sandbox; (2) formal transposition of the exaptation/spandrel/multilevel selection triad to

legal theory; (3) four new metrics (FDI, SEM, MSR, MCI) for norm evolution tracking; (4) a PoC on Argentine labor reform with four falsifiable predictions (Section 4.2).

## 2. Theoretical Framework

### 2.1 Exaptation and Current Function (Gould & Vrba, 1982)

Gould and Vrba introduced exaptation to describe features co-opted from structures serving a different original function, or arising as byproducts later recruited into service. The transposition to normative analysis is direct. Legal norms have an origin function — declared in legislative intent — and a current function, that is, the actual functional role the norm plays in ongoing institutional dynamics. As Nietzsche argued (cited approvingly by Gould): "the cause of the origin of a thing and its eventual utility lie worlds apart" (Zur Genealogie der Moral, §11.12). Dawkins' (1982) extended phenotype concept deepens this: norms are phenotypic extensions of institutional agents' dispositions, subject to the same exaptive dynamics as biological traits.

In Argentine labor law, the indemnización por despido sin causa (1974) was designed to protect workers from arbitrary termination. Its current function — in a market with 37–40% informality — is increasingly a negotiation instrument. SNMS tracks this drift through the Functional Drift Index (FDI):

$$\text{FDI}(t) = 1 - \text{cos\_similarity}(\text{origin\_vector}(N), \text{usage\_vector}(N, t))$$

FDI approaching 0: stable alignment between declared and observed function. FDI approaching 1: full exaptation. The PoC achieves FDI = 0.42 by round 20.

### 2.2 Spandrels (Gould & Lewontin, 1979)

Gould and Lewontin's spandrels of San Marco paper argues that byproducts of structural constraints are real, common, and consequential — and that treating every feature as a selected adaptation misrepresents reality. Normative spandrels are emergent behavioral dynamics arising as byproducts of norm interactions that subsequently acquire autonomous functional significance. In the Argentine context: strategic pre-dismissal documentation preparation — not legislated, not in any norm's action set — is a spandrel of the interaction between high indemnity levels, judicial variability, and asymmetric information. SNMS detects them through the Spandrel Emergence Metric (SEM):

$$\text{SEM}(A', t) = |P(A'|N, t) - P(A'|\neg N, t)| \times \text{stability\_coefficient}(A', t)$$

### 2.3 Multilevel Selection (Wilson & Sober, 1994; Okasha, 2006)

SNMS operates three selection levels: L1 (individual compliance), L2 (archetype-group jurisprudential norms), L3 (system-level normative stability). The Multilevel Selection Ratio (MSR):

$$\text{MSR} = \text{Delta\_group} / (\text{Delta\_group} + \text{Delta\_individual})$$

MSR > 0.5: group-level selection dominant. MSR < 0.5: individual-level dominant. The PoC yields MSR = 0.73 vs. 0.49 in the control — not a difference of magnitude but a change of regime: the norm is maintained by archetype-group dynamics, not by individual incentives. This has direct implications for reform design.

### 3. SNMS Architecture

#### 3.1 MindProfiles

Each agent holds a MindProfile with three dimensions: CognitiveDimension (processing\_capacity, processing\_load, bayesian\_plasticity, working\_memory\_depth); AffectiveDimension (institutional\_trust, risk\_aversion, loss\_aversion\_coefficient); DoctrinalDimension (interpretive\_school, stare\_decisis\_weight, intertextual\_breadth). Each archetype is assigned philosophical anchors as calibration priors for the doctrinal and affective parameters.

#### 3.2 TribeV2 Integration: Architecture and Current State

Designed architecture (full integration, future work): at each simulation round, the active norm's text excerpt passes through TribeV2's language encoder. The resulting cortical prediction vector updates each archetype's processing\_load:

```
processing_load(k, t) = alpha × TribeV2_load_signal(text(N)) + (1-alpha) ×
baseline_load(k)
```

Current PoC state: the processing\_load parameters are assigned manually from the theoretical and empirical literature on cognitive differentiation across institutional roles. The architecture is fully compatible with TribeV2 — the MindProfile structure accepts a tribe\_encoding vector — but TribeV2 does not run inside the PoC. The TribeV2-derived values in Section 4 are theoretically motivated estimates, not outputs of the encoder. Full API integration is the primary next step.

Note on repository state: minds/archetypes/, agents/, engine/, norms/, and analysis/ are implemented. The tribe\_bridge.py module (minds/tribe\_bridge.py) and sandbox/runner.py are architectural stubs pending TribeV2 API access. The PoC runs end-to-end without TribeV2 using manually calibrated parameters.

#### 3.3 Council Bridge, Effective CRI, and HBU

High-stakes decisions route through a deliberative Council Bridge activating the agent's philosophical anchor pair. Each agent computes an effective CRI:

```
effective_CRI = CRI_base × effective_cri_modifier()
```

For the JuezCSJN:  $0.82 \times 1.30 = 1.07$  (capped at 1.0). The modifier of 1.30 arises from high institutional\_trust (+0.25), Stoic mentor anchor (+0.10), and high stare\_decisis\_weight (+0.08), offset by moderate risk\_aversion (−0.13). Beliefs update via Hierarchical Belief Update (HBU) with rate  $\beta = \text{bayesian\_plasticity}$  (JuezCSJN: 0.38; DirigenteSindical: 0.71).

### 4. Proof of Concept: Argentine Labor Reform

#### 4.1 Experimental Design

100 agents, 8 archetypes, 20 rounds (~45 institutional days each), 5 random seeds, 3 active norms (N1: dismissal indemnity; N2: collective agreement reform; N3: dismissal fund alternative). The full experiment (future work) will use 500 agents and 50 seeds.

Table 1. Archetype Distribution and MindProfile Anchors

| Archetype             | N  | CRI base | Anchors             | proc.cap. | inst.trust |
|-----------------------|----|----------|---------------------|-----------|------------|
| JuezCSJN              | 8  | 0.82     | Aristotle, Aurelius | 0.91      | 0.85       |
| DirigenteSindical     | 12 | 0.67     | Gramsci, Rawls      | 0.63      | 0.42       |
| EmpresarioPyme        | 15 | 0.54     | Hayek, Schumpeter   | 0.55      | 0.38       |
| TrabajadorFormal      | 20 | 0.71     | Rousseau, Rawls     | 0.52      | 0.61       |
| TrabajadorInformal    | 18 | 0.43     | Hobbes, Nozick      | 0.39      | 0.21       |
| LegisladorOposicion   | 10 | 0.61     | Montesquieu, Berlin | 0.68      | 0.55       |
| CiudadanoInformado    | 12 | 0.58     | Kant, Mill          | 0.58      | 0.49       |
| TecnocrataMinisterial | 5  | 0.76     | Weber, Popper       | 0.81      | 0.72       |

## 4.2 Falsifiable Predictions

P1 (Compliance Legitimacy Score): CLS under SNMS with MindProfiles will exceed 0.85 in the final round, driven by the structurally anchored JuezCSJN and TecnocrataMinisterial archetypes. This prediction would be falsified by  $CLS < 0.85$  at round 20 across all 5 seeds.

P2 (Functional Drift): FDI for norm N1 (dismissal indemnity) will exceed 0.40 by round 15, indicating that the norm's current function diverges substantially from its declared origin function. Falsified by  $FDI(N1, t=15) < 0.40$ .

P3 (Spandrel emergence): At least one normative spandrel — a stable behavioral pattern not in any norm's original action set — will emerge by round 15, arising from the interaction of N1 and N3. Falsified by zero confirmed spandrels (SEM threshold not crossed) before round 15.

P4 (Selection regime): MSR will exceed 0.65, confirming that group-level selection at the archetype level is the primary mechanism of norm maintenance. Falsified by  $MSR < 0.65$ . This would indicate that individual incentive-based reform strategies are sufficient — the opposite of the SNMS prediction.

## 4.3 Results

Table 2. Key Metrics by Round (means across 5 seeds; CLS = MCI weighted by doctrinal legitimacy scores)

| Round | MCI  | FDI (N1) | FDI (N3) | MSR  | Spandrels |
|-------|------|----------|----------|------|-----------|
| 1     | 0.91 | 0.08     | —        | —    | 0         |
| 5     | 0.87 | 0.17     | 0.09     | 0.64 | 0         |
| 10    | 0.84 | 0.29     | 0.21     | 0.71 | 1         |
| 14    | 0.81 | 0.38     | 0.31     | 0.74 | 2         |
| 20    | 0.89 | 0.42     | 0.36     | 0.73 | 2         |

All four predictions confirmed. The CLS of 0.89 at round 20 is higher than raw MCI because the complying agents at equilibrium are disproportionately the high-legitimacy archetypes (judges, technocrats). The MSR of 0.73 vs. 0.49 in the control is a change of regime, not of magnitude.

#### 4.4 Emergent Spandrels

Spandrel S1 — Pre-dismissal documentation preparedness (round 10, SEM = 0.67): EmpresarioPyme develops systematic documentation accumulation prior to any dismissal event as a byproduct of N1's indemnity structure interacting with N3's requirements. By round 14, 84% of EmpresarioPyme and 23% of TecnocrataMinisterial agents have adopted this pattern. The LITIGATE action has been exapted from a terminal dispute-resolution event into a permanent structural feature of the employment relationship — the most direct empirical illustration of the Gould–Nietzsche thesis in the normative domain.

Spandrel S2 — Strategic coalition for collective agreement bypass (round 14, SEM = 0.54): DirigenteSindical and LegisladorOposicion develop coordinated action patterns not individually rational for either, synchronizing legislative delay with collective action to block N2. This coalition space opened as a byproduct of N1–N2 interaction and was subsequently 'decorated' — precisely as the San Marco spandrels were decorated once the structural space existed.

#### 4.5 Control Comparison

Control simulation (same 100 agents, no MindProfiles, static compliance scalars matched to CRI\_base): MCI = 0.83, FDI(N1) = 0.28, MSR = 0.49, zero spandrels at round 20. The spandrel result is categorical: parameterized automata produce no emergent behavioral patterns because they lack any mechanism for inter-archetype behavioral transmission.

## 5. Discussion

### 5.1 Structural Effect of Cognitive MindProfiles

The most consequential result is not the CLS level but the mechanism sustaining it. The JuezCSJN's effective CRI of 1.07 (capped at 1.0) means that even under active defection by peripheral archetypes, the judicial archetype maintains near-perfect compliance and, through the HBU social modifier, exerts upward pressure on proximate archetypes. The Aristotle anchor provides deliberative rationality (high stare\_decisis\_weight); Marcus Aurelius provides Stoic institutional commitment (slow decay under adversarial conditions). Together they create a compliance anchor no parameterized scalar can replicate.

### 5.2 The LITIGATE Spandrel as Functional Exaptation

Spandrel S1 is simultaneously a spandrel and an exaptation: it arose from norm interaction (spandrel) and was then recruited into a distinct functional role (exaptation) — not merely defensive but actively constitutive of the employment relationship itself. The LITIGATE action, originally a terminal event, now serves a function entirely distinct from its origin. The utility of the norm is not its origin.

### 5.3 Limitations and Current State of Implementation

Three limitations require explicit acknowledgment. First, TribeV2 does not run inside the PoC. The `processing_load` values are theoretically motivated estimates from the empirical literature on cognitive differentiation; they are inspired by TribeV2's architecture but are not its outputs. This is the primary validation gap. Second, even once TribeV2 is integrated, it will provide population-average brain predictions, not archetype-specific neural activity. The zero-shot generalization capability makes the proxy defensible, but the gap between average prediction and archetype-specific processing remains a source of model error. Third, the PoC uses 5 seeds and 100 agents; preliminary sensitivity analyses suggest MSR estimates may shift  $\pm 0.08$  with larger populations.

## 6. Conclusion

SNMS represents a substantive departure from first-generation normative simulation. By replacing parameterized automata with cognitively differentiated agents, SNMS generates emergent norm dynamics that parameterized models cannot produce: spandrels, exaptive functional drift, and group-level selection regimes that resist individual-incentive reform strategies. The PoC results — CLS 0.89, FDI 0.42, MSR 0.73, two emergent spandrels — establish proof of feasibility and confirm four falsifiable predictions.

The integration of TribeV2 as a cognitive load proxy opens a methodological pathway not previously explored: using a brain foundation model to drive differential cognitive processing in a legal simulation environment. The immediate agenda is to replace the current manual calibration with actual TribeV2 encoder outputs over Argentine normative text corpora. That step will determine whether the MindProfile architecture, already producing consequential results with theoretical parameters, performs even more precisely when grounded in empirical neural encoding.

The mind, properly modeled, makes the judge.



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